

DIPTARAJ SINHA

[My Portfolio](#)

- Kolkata, West Bengal • diptarajsinha1014@gmail.com • +91 91635 31884
- LinkedIn: [linkedin.com/in/diptaraj-sinha](https://www.linkedin.com/in/diptaraj-sinha) • GitHub: github.com/DiptarajSinha

PROFESSIONAL SUMMARY

AI Developer and Front-End Engineer delivering production-ready, AI-powered web solutions. Built LLM-integrated chat systems and intelligent interfaces using React and modern JavaScript. Engineered cloud-based architectures on AWS with secure authentication and REST API integration.

Improved model inference reliability, reduced latency, and optimized infrastructure utilization. Seeking an AI Engineer / Front-End role to develop scalable, data-driven digital products.

TECHNICAL SKILLS

- Languages: Python, Java, JavaScript, C++, C
- Frontend: React, HTML, CSS
- Backend: SQL, MySQL, FastAPI, REST APIs
- Cloud: AWS (EC2, S3, Lambda, IAM), Docker (basic)
- ML/AI: Prompt Engineering, scikit-learn

WORK EXPERIENCE

Generative AI Developer Intern — AI WALLAH | Jun 2025 – Jul 2025

- Integrated LLM-powered features into React applications, boosting chatbot response relevance by 35% through targeted prompt engineering.
- Designed prompt templates and caching layers to reduce API calls by 25%, lowering monthly costs by ₹900.
- Built context-aware conversation flows, improving multi-turn accuracy by 20% in testing.

AWS Cloud Security Intern — Employability.life | Apr 2025 – Jun 2025

- Developed and executed 25+ test cases for AWS EC2, S3, and Lambda workflows, identifying 12 security gaps and leading to 100% remediation.
- Implemented IAM policy simulations, reducing over-privileged roles by 40%.
- Collaborated with engineering to harden deployments, achieving zero high-severity findings in post-audit reviews.

PROJECTS

- **My portfolio website** : [View Demo](#) - Performance-optimized developer portfolio built with Next.js, featuring high-fidelity interactive UI and responsive design.
- **The Ultron** : [View Demo](#) - Autonomous robotic tour system combining an ESP32 robot with a Next.js platform, featuring AI-driven booking, automated payments, and real-time synchronization.
- **TransitPay** : [Android APK](#) | [Github](#) - Engineered a high-performance, full-stack digital transit platform using Flutter and Supabase, featuring real-time QR verification, Stripe payment integration, and encrypted offline-first data synchronization.
- **RAG-Powered QA System** : [View Demo](#) - Engineered a high-performance RAG pipeline using FastAPI, Gemini 2.5 Flash, and Supabase (pgvector) to provide accurate, cloud-persistent semantic search and intelligent querying for large PDF datasets
- **AI Weather Hub** : [View Demo](#) - Real-time weather app with AI assistance (React + Gemini).
- **Chat with U.L.T.R.O.N** : [View Demo](#) - An AI chatbot powered by Gemini.
- **Personal Dev Dashboard** : [View Demo](#) - A centralized development monitoring tool built with TypeScript and JavaScript, featuring real-time system tracking and a high-performance CSS layout.
- **Trace** : [View Demo](#) - A real-time behavioral analytics dashboard built with React that utilizes heuristic analysis to classify and visualize visitor intent as Human, Bot, or Threat.
- **Drafted** : [View Demo](#) - A high-performance digital library built with HTML, CSS, and JS, featuring a Fluent UI-driven interface for an immersive book reading experience.

- **Network Anomaly Detector** : [View Demo](#) - An AI-driven monitoring system that utilizes machine learning to identify and alert on irregular network traffic patterns and potential security threats.
- **ScreenTest** : [NPM Package](#) - An automated visual regression testing framework featuring high-fidelity pixel comparison and styled reporting to ensure UI consistency across deployments.
- **Reddit Scraper** : [Github](#) - A behavioral profiling tool developed in Python that utilizes the Gemini API to scrape and analyze Reddit data for automated user persona generation.
- **Voice-Flix** : [View Demo](#) - A hands-free web application developed with HTML, CSS, and JS, featuring a custom voice-command architecture for total interactive control without physical input.
- **Quiz-o-pedia** : [Github](#) - A full-stack AI-driven assessment platform engineered with Next.js and NestJS, featuring automated question generation and an optimized database architecture for real-time leaderboards and user analytics.
- **Credit Card Fraud Detection** : [Github](#) - A machine learning solution developed in Python that utilizes Logistic Regression and SMOTE (Synthetic Minority Over-sampling Technique) to identify fraudulent transactions within highly imbalanced financial datasets.
- **Digital Memory Palace** : [View Demo](#) - An immersive 3D productivity environment developed with React and Three.js for spatial organization and interactive visualization of complex data.
- **Neural Dream Visualizer** : [View Demo](#) - An experimental generative art platform engineered with React and Vite, featuring high-performance rendering of AI-driven dreamscapes and fluid visual narratives.
- **Quantum UI States** : [View Demo](#) - An experimental interface framework engineered with React and Vite, utilizing non-linear state logic to render UI elements that exist in multiple interactive states simultaneously.
- **Liquid Interface** : [View Demo](#) - A fluid design system developed with HTML and CSS, utilizing advanced physics-based transitions to create UI components that dynamically flow and adapt to user interactions.
- **Predictive Interface** : [View Demo](#) - An intelligent UI framework developed with HTML and JS that utilizes behavioral anticipation logic and pre-emptive rendering to adapt interface elements based on predicted user actions.
- **Emotion-Responsive UI** : [View Demo](#) - A specialized interface framework developed with HTML, CSS, and JS that utilizes biometric data and sentiment analysis to dynamically adapt visual elements based on real-time user emotional states.
- **Reverse Scroll Navigation** : [View Demo](#) - A non-linear navigation framework developed with HTML, CSS, and JS that implements an inverted scroll architecture and chronologically-sequenced animations to simulate a temporal reverse-scrolling experience.
- **Glitch Art Generator** : [Github](#) - A digital media tool developed in Python that utilizes algorithmic image manipulation and data-bending techniques to programmatically generate aesthetic glitch art from raw image files.

EDUCATION

B.Tech, Computer Science & Engineering — Sister Nivedita University

Expected May 2026 • CGPA (up to 6th semester): 8.63

CERTIFICATIONS

- **IBM** : AI Developer (June 2025) [Certificate](#)
- **NVIDIA** : Building LLM Applications With Prompt Engineering (June 2025) [Certificate](#)
- **Google** : IT Automation with Python (June 2025) [Certificate](#)
- **IBM** : Data Analysis with Python (June 2025) [Certificate](#)
- **University of Michigan** : Applied ML in Python (May 2025) [Certificate](#)
- **University of Colorado Boulder** : SQL (April 2025) [Certificate](#)